Syed Muhammad Abdullah Shah

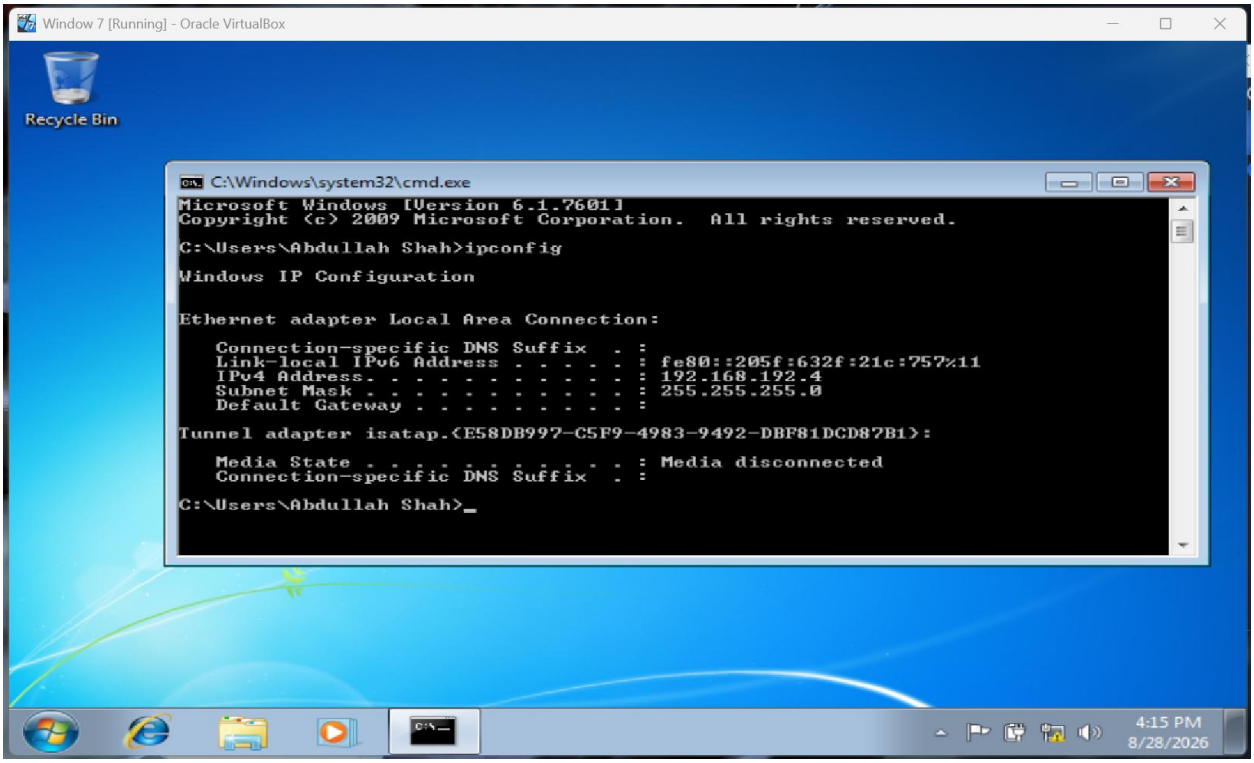
Roll No: HM-B2-2026-1037-296

Q1: Authorized Network Reconnaissance & Traffic Analysis Lab

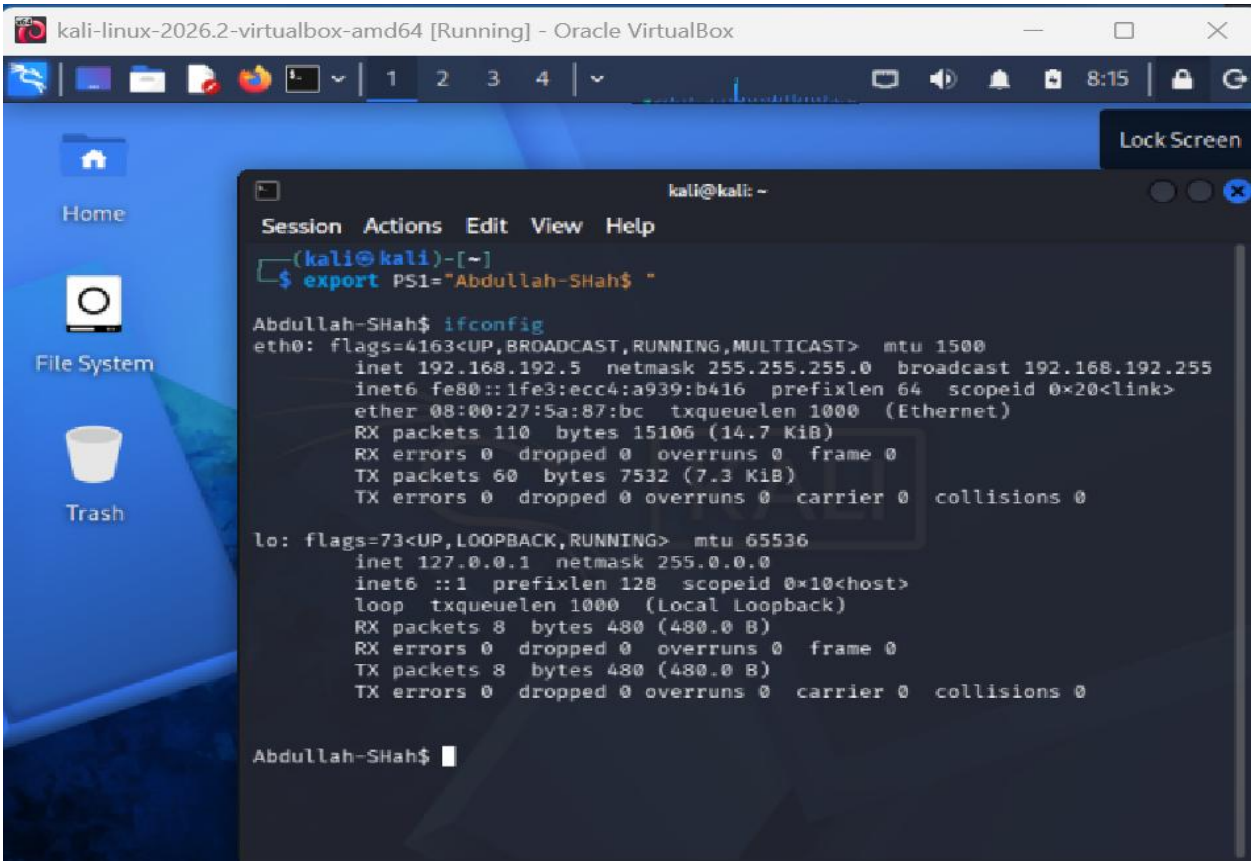
1. Lab Setup & Connectivity

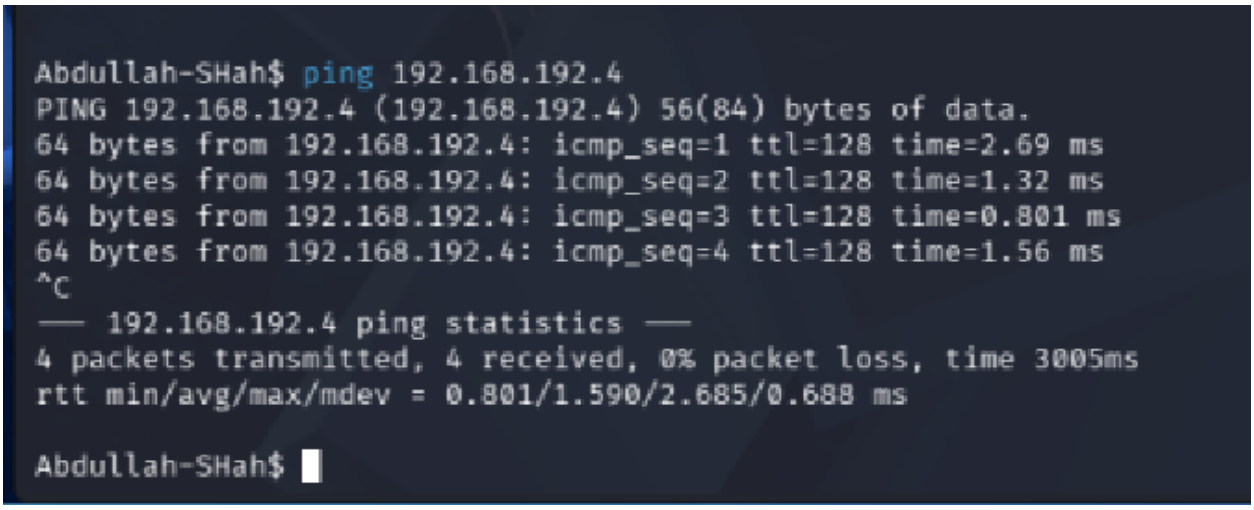
There were 2 Virtual Machines

- **Windows 7 (Target Machine):** The IP address is 192.168.192.4.
- **Kali Linux (Attacker/Auditor Machine):** The IP address is 192.168.192.5.
- Verifying the IPv4 address (192.168.192.4) of the Windows 7 target machine using the ipconfig command.



Checking the network interface and IPv4 address (192.168.192.5) of the Kali Linux machine using ifconfig.



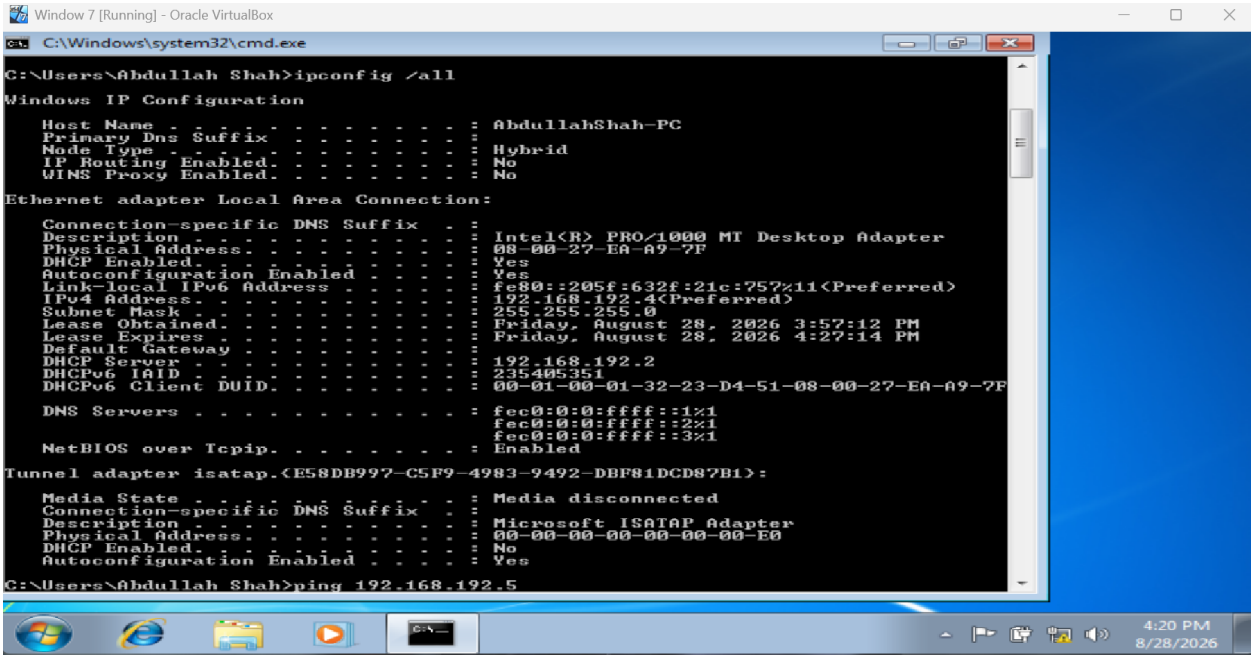


Successful ICMP ping test from Kali Linux to Windows 7, verifying two-way network connectivity

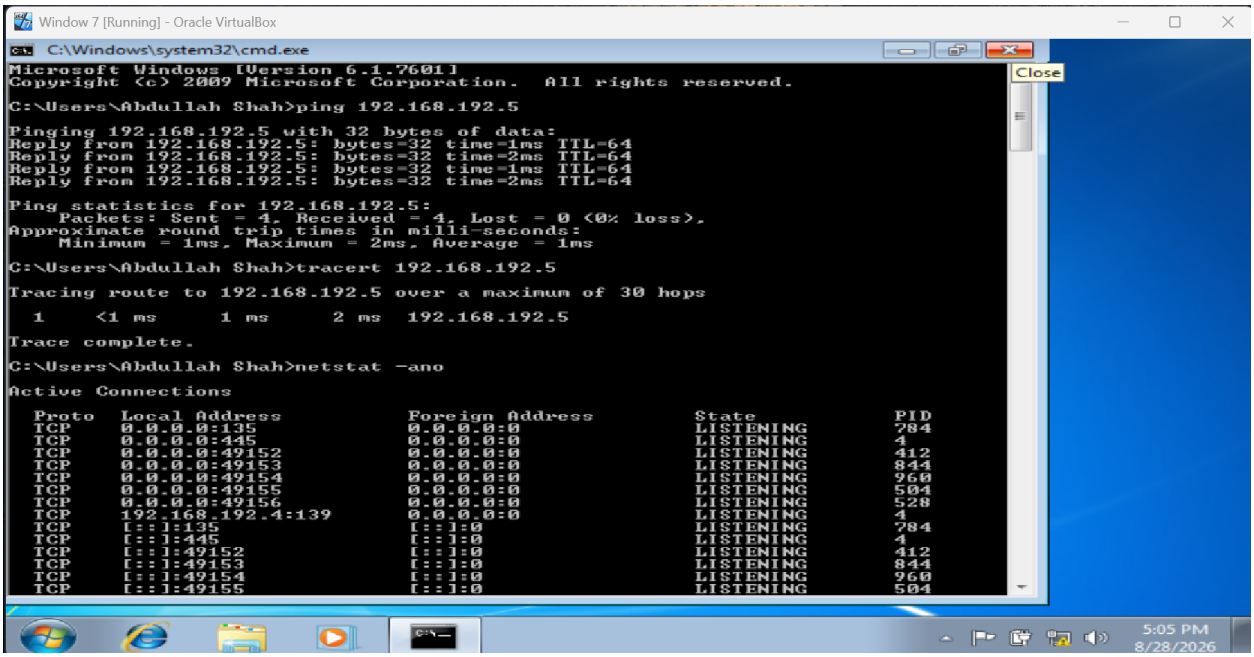
2. CLI & Linux Commands Demonstration

Firstly, **Windows Commands** ipconfig /all was used to view detailed network adapter information including MAC addresses. netstat -ano displayed active listening ports and their associated PIDs.

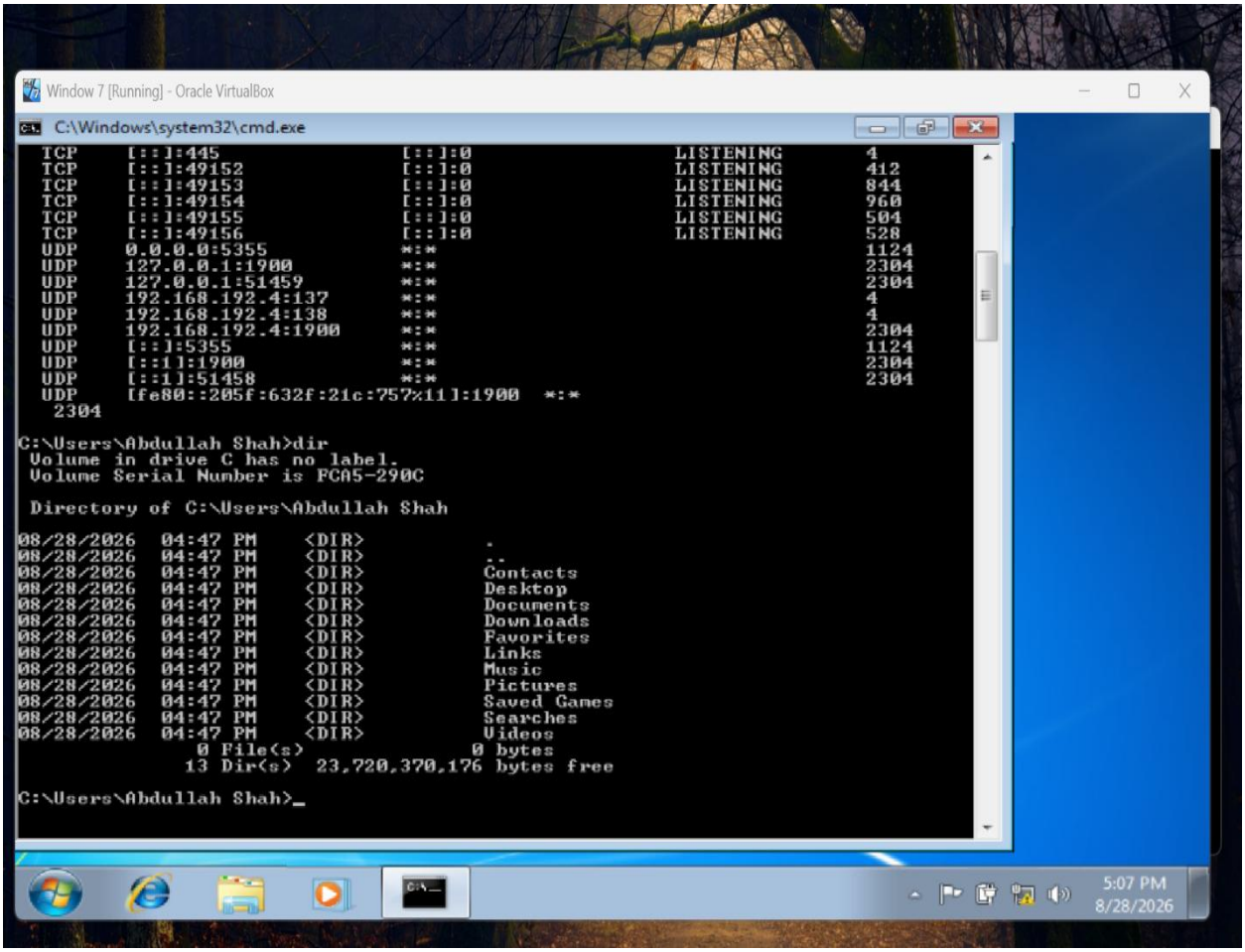
The dir command was used to list directory contents and to verify the file system structure. tracert was used to map the network path to the target.



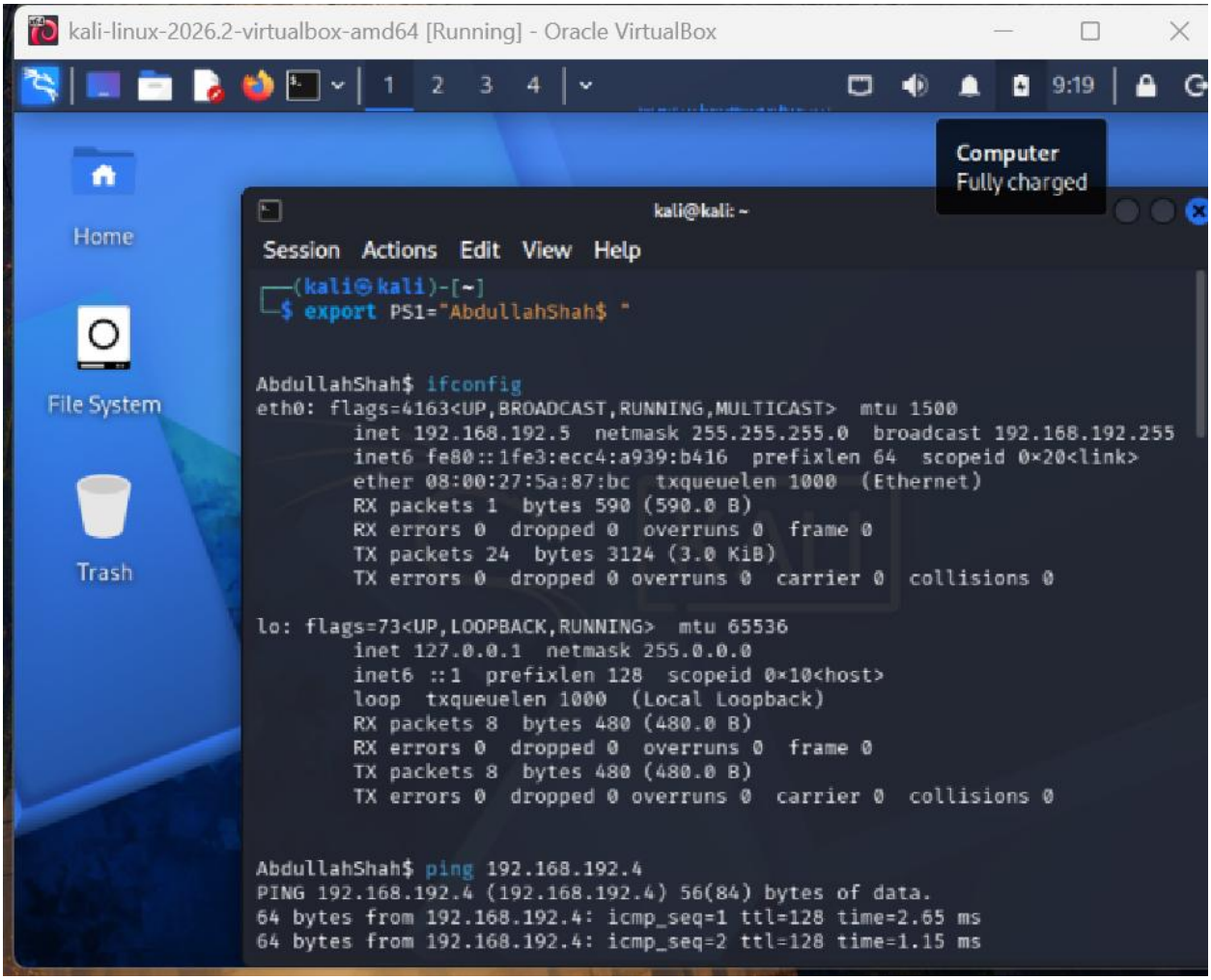
Executing ipconfig /all, ping, tracert, and netstat -ano on Windows 7 to gather detailed network and active connection data.



Using the dir command to list the contents of the current directory in Windows 7.

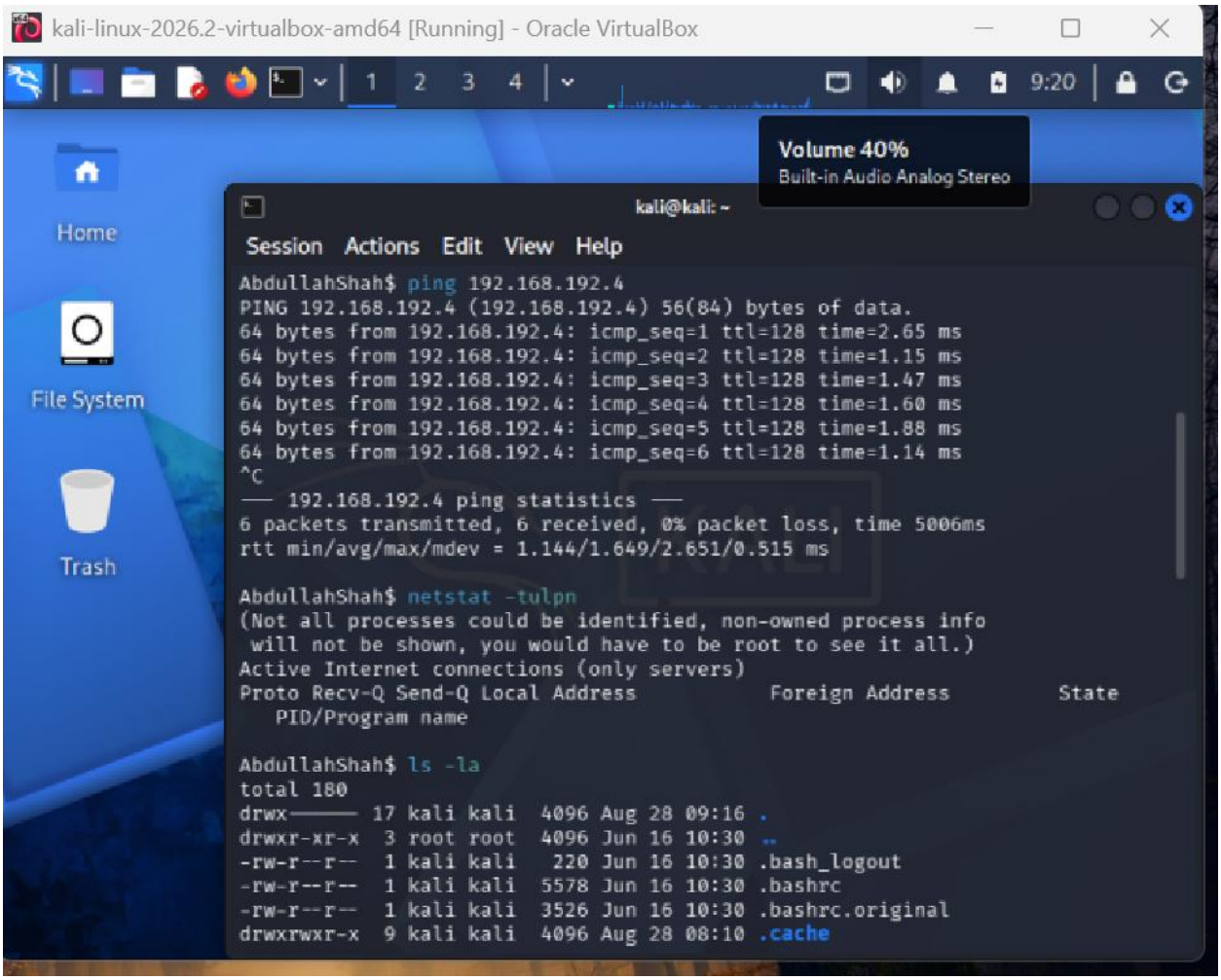


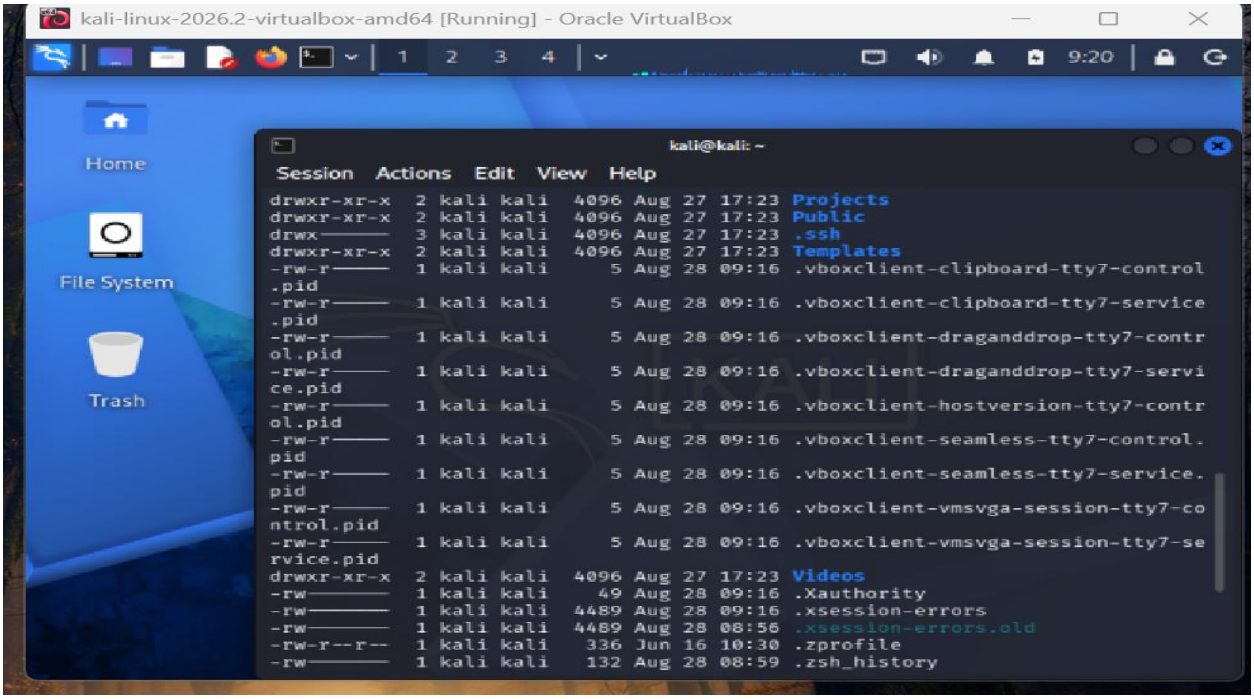
Now moving towards **Linux Commands,** pwd printed the current working directory (/home/kali). ls -la listed all files including hidden files with their permissions. cat /etc/os-release was used to verify the OS version, confirming it as Kali Linux. Moreover, we check the ip through ifconfig and ping command to verify connectivity.

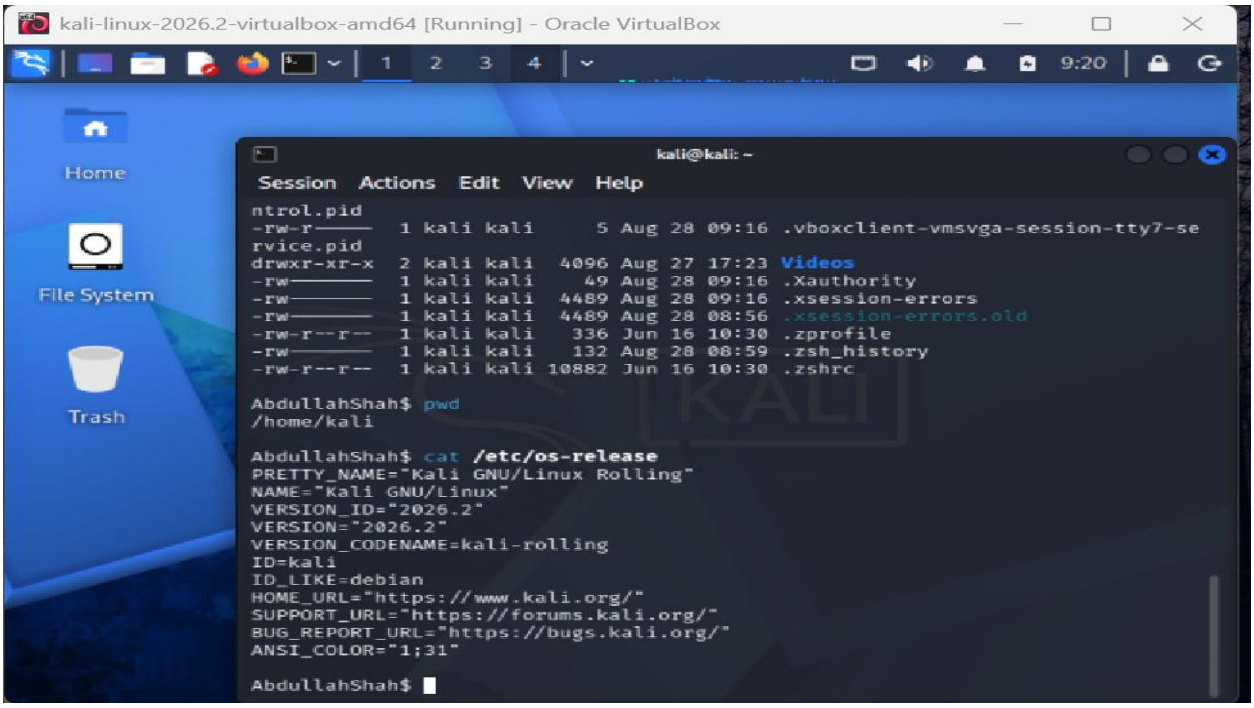


Utilizing Linux CLI commands (ls -la, pwd, cat /etc/os-release) to navigate directories, view file permissions, and verify the operating system details.

Using netstat -tulpn to check active connections, listening ports, and network statistics.


kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
1 2 3 4
9:20
Volume 40%
Built-in Audio Analog Stereo
Home
File System
Trash
kali@kali: ~
Session Actions Edit View Help
AbdullahShah$ ping 192.168.192.4
PING 192.168.192.4 (192.168.192.4) 56(84) bytes of data.
64 bytes from 192.168.192.4: icmp_seq=1 ttl=128 time=2.65 ms
64 bytes from 192.168.192.4: icmp_seq=2 ttl=128 time=1.15 ms
64 bytes from 192.168.192.4: icmp_seq=3 ttl=128 time=1.47 ms
64 bytes from 192.168.192.4: icmp_seq=4 ttl=128 time=1.60 ms
64 bytes from 192.168.192.4: icmp_seq=5 ttl=128 time=1.88 ms
64 bytes from 192.168.192.4: icmp_seq=6 ttl=128 time=1.14 ms
^C
— 192.168.192.4 ping statistics —
6 packets transmitted, 6 received, 0% packet loss, time 5006ms
rtt min/avg/max/mdev = 1.144/1.649/2.651/0.515 ms
AbdullahShah$ netstat -tulpn
(Not all processes could be identified, non-owned process info
will not be shown, you would have to be root to see it all.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address Foreign Address State
PID/Program name
AbdullahShah$ ls -la
total 180
drwx------ 17 kali kali 4096 Aug 28 09:16 .
drwxr-xr-x 3 root root 4096 Jun 16 10:30 ..
-rw-r--r-- 1 kali kali 220 Jun 16 10:30 .bash_logout
-rw-r--r-- 1 kali kali 5578 Jun 16 10:30 .bashrc
-rw-r--r-- 1 kali kali 3526 Jun 16 10:30 .bashrc.original
drwxrwxr-x 9 kali kali 4096 Aug 28 08:10 .cache


kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
1 2 3 4
9:20
Home
File System
Trash
kali@kali: ~
Session Actions Edit View Help
drwxr-xr-x 2 kali kali 4096 Aug 27 17:23 Projects
drwxr-xr-x 2 kali kali 4096 Aug 27 17:23 Public
drwx------ 3 kali kali 4096 Aug 27 17:23 .ssh
drwxr-xr-x 2 kali kali 4096 Aug 27 17:23 Templates
-rw-r----- 1 kali kali 5 Aug 28 09:16 .vboxclient-clipboard-tty7-control.pid
-rw-r----- 1 kali kali 5 Aug 28 09:16 .vboxclient-clipboard-tty7-service.pid
-rw-r----- 1 kali kali 5 Aug 28 09:16 .vboxclient-draganddrop-tty7-control.pid
-rw-r----- 1 kali kali 5 Aug 28 09:16 .vboxclient-draganddrop-tty7-service.pid
-rw-r----- 1 kali kali 5 Aug 28 09:16 .vboxclient-hostversion-tty7-control.pid
-rw-r----- 1 kali kali 5 Aug 28 09:16 .vboxclient-seamless-tty7-control.pid
-rw-r----- 1 kali kali 5 Aug 28 09:16 .vboxclient-seamless-tty7-service.pid
-rw-r----- 1 kali kali 5 Aug 28 09:16 .vboxclient-vmsvga-session-tty7-control.pid
-rw-r----- 1 kali kali 5 Aug 28 09:16 .vboxclient-vmsvga-session-tty7-service.pid
drwxr-xr-x 2 kali kali 4096 Aug 27 17:23 Videos
-rw------- 1 kali kali 49 Aug 28 09:16 .Xauthority
-rw------- 1 kali kali 4489 Aug 28 09:16 .xsession-errors
-rw------- 1 kali kali 4489 Aug 28 08:56 .xsession-errors.old
-rw-r--r-- 1 kali kali 336 Jun 16 10:30 .zprofile
-rw------- 1 kali kali 132 Aug 28 08:59 .zsh_history



3. Network Concepts

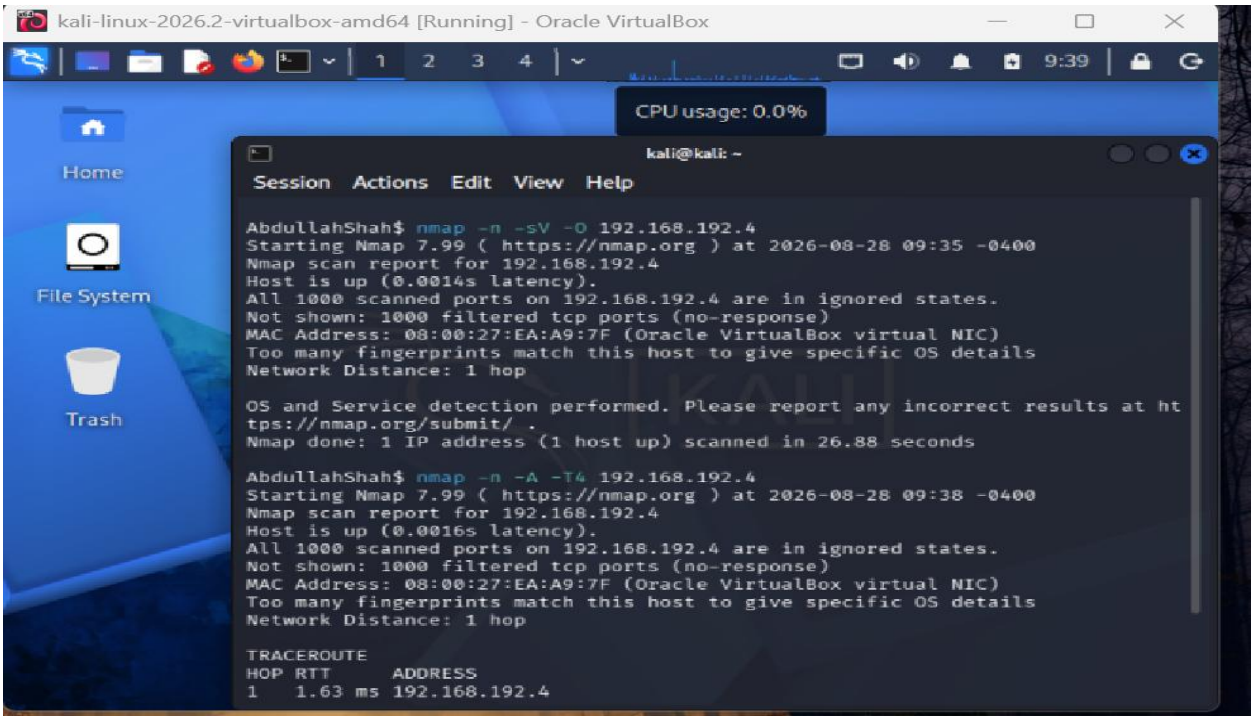
The communication between the Windows 7 and Kali Linux virtual machines was with the help of the TCP/IP network model.

- **Layer 2 (Data Link Layer):** Both machines communicate using MAC addresses (e.g., 08:00:27:EA:A9:7F). Address Resolution Protocol (ARP) is used to map IP addresses of these physical MAC addresses.
- **Layer 3 (Network Layer):** The systems use IPv4 addresses (192.168.192.4 and 192.168.192.5) to route packets between each other within the isolated subnet (255.255.255.0)
- **Layer 4 (Transport Layer):** This layer manages how data is sent between the two machines. TCP is used for reliable, connection-based communication (as seen in the port scanning results), while ICMP (used for ping) works directly at the Network layer.
- **Layer 7 (Application Layer):** This is the top layer where user commands like ping, nmap, and file sharing requests are generated before being passed down through the other layers for transmission.

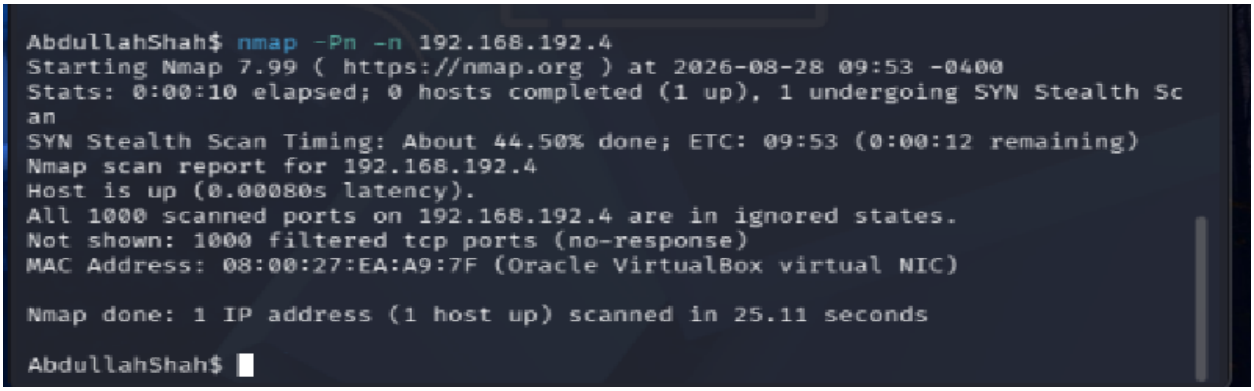
4. Network Discovery & Information Gathering

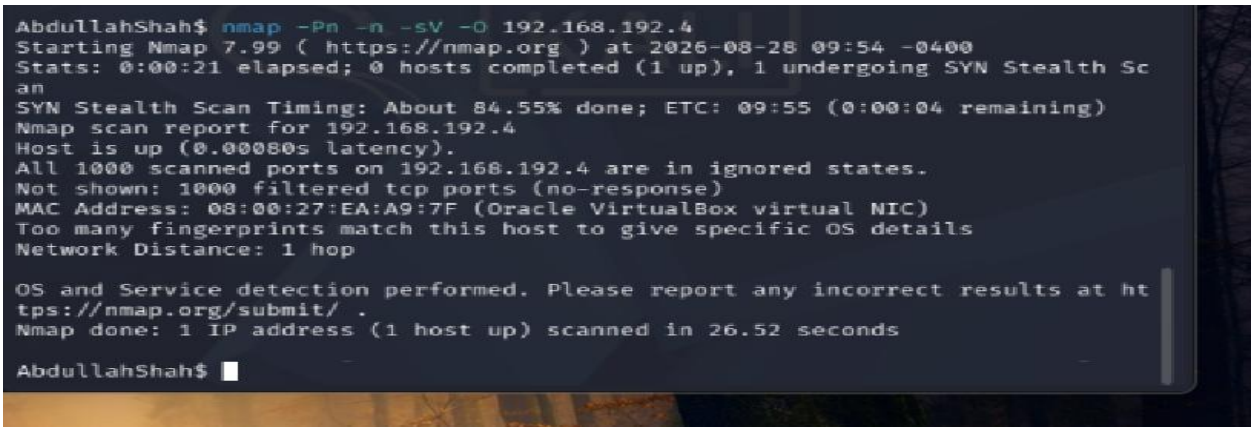
Nmap CLI: Command such as -sV (service version detection) and -O (OS detection) were used. The scans showed ports 135/tcp, 139/tcp, and 445/tcp are open.

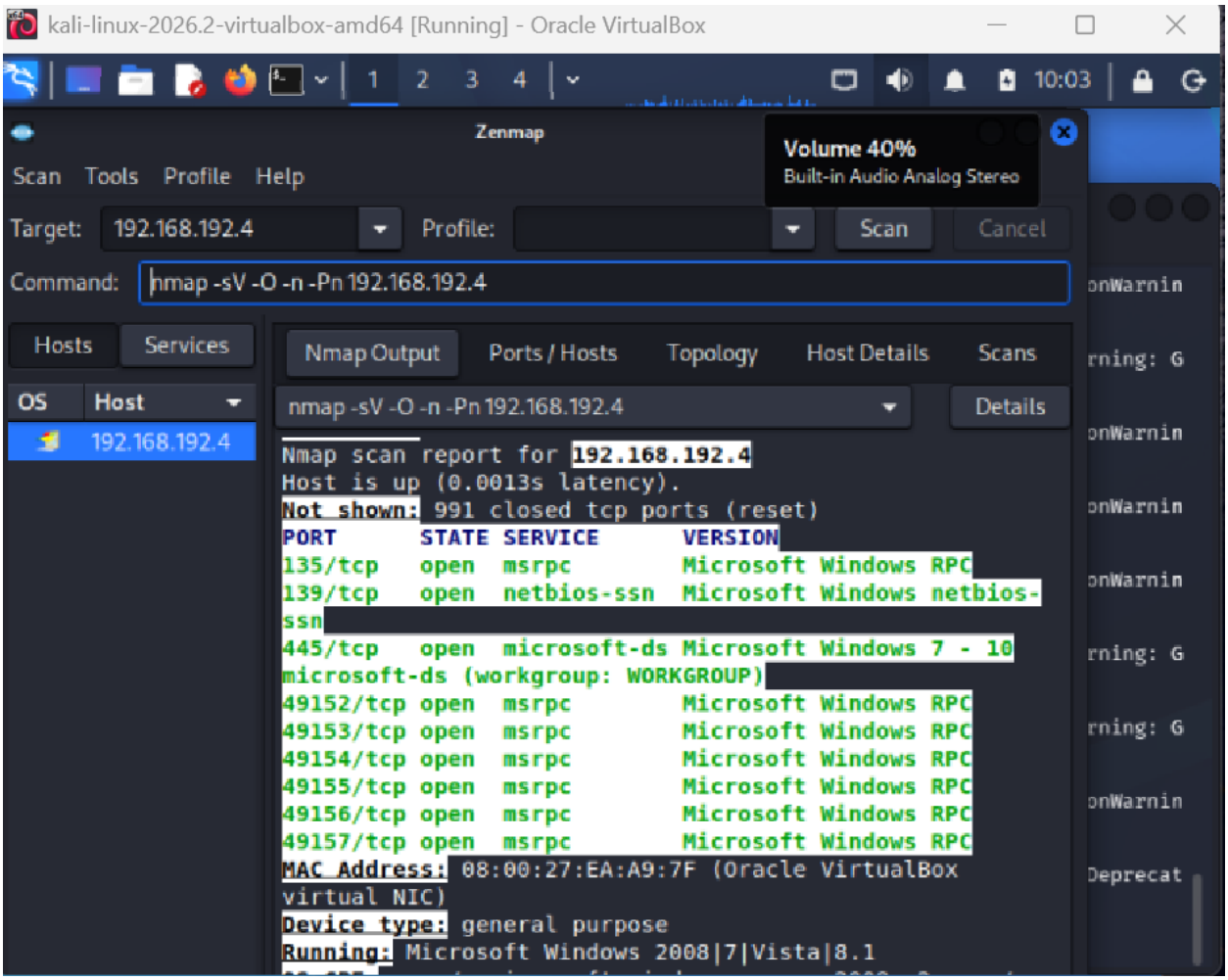
Zenmap GUI: Zenmap provided a visual topology and detailed host status. It identified the operating system as Microsoft Windows 7 / Vista / Server 2008 and detected services like msrpc and microsoft-ds.



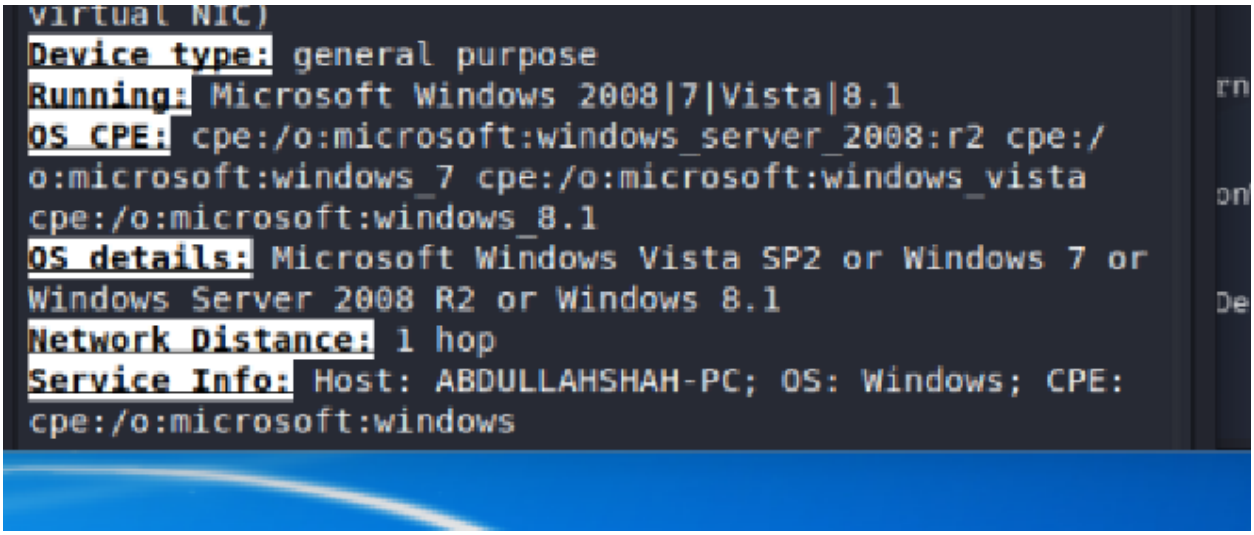
Nmap terminal execution showing host discovery, OS detection, and port scanning results against the target IP.



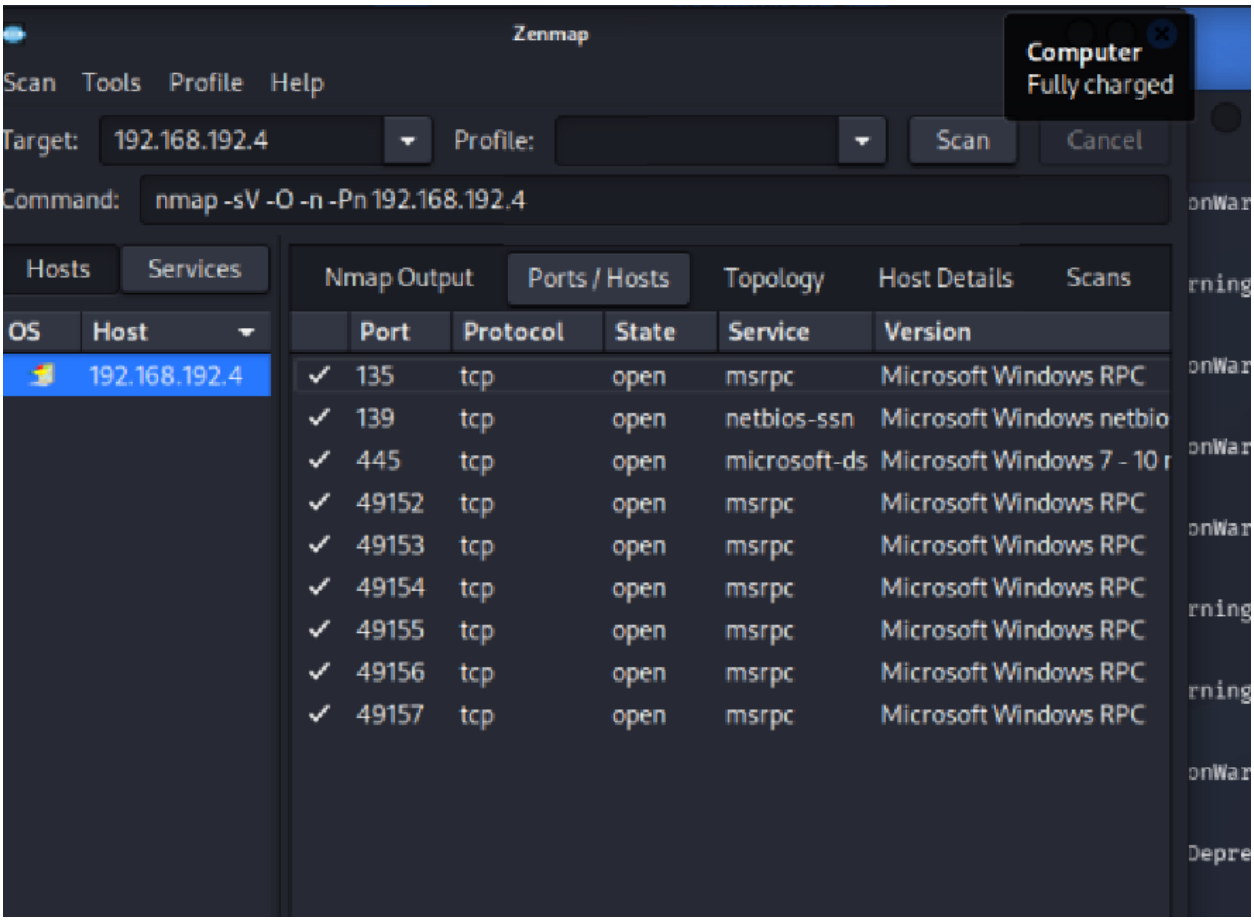


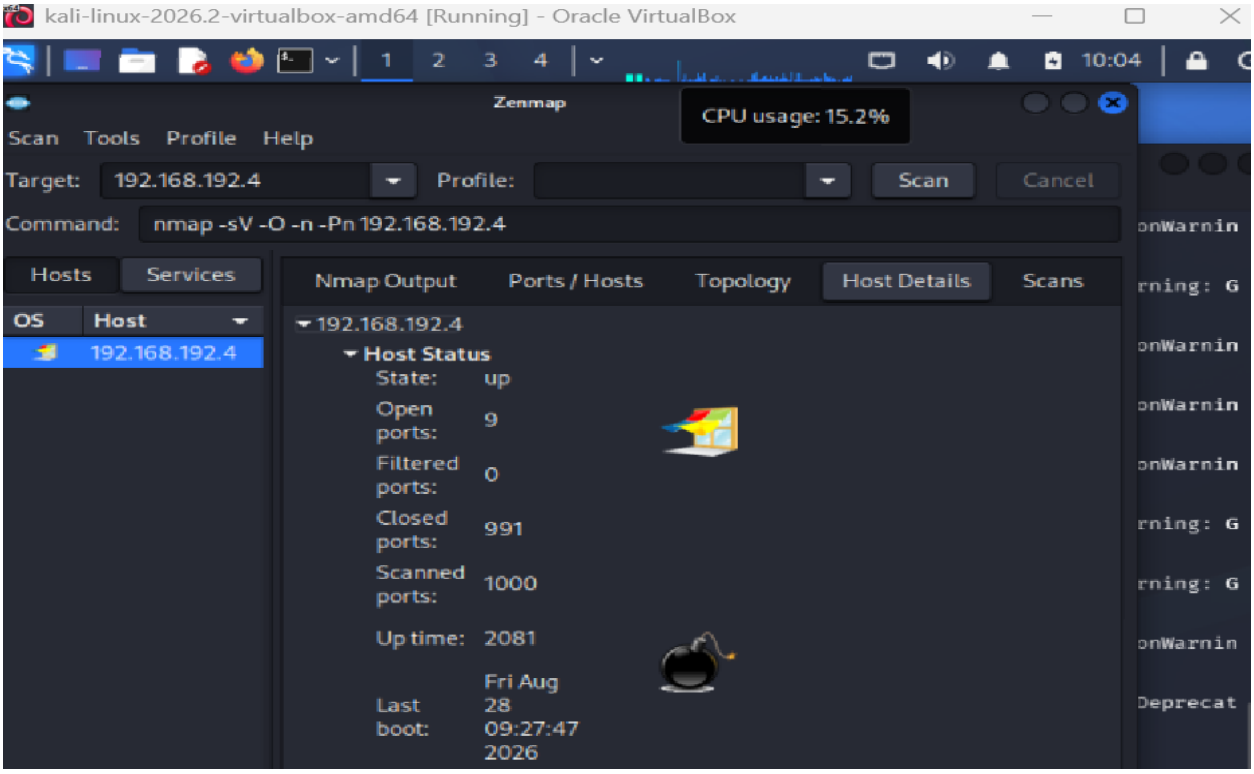


Information Gathering: Basic information gathering wasdone by identifying the services running on the target machine through Nmap's service version detection . This revealed the exact service names (microsoft-ds, msrpc) running on the open ports, which helped in understanding what software the target machine was running without directly accessing it.



Zenmap GUI displaying the list of open ports (135, 139, 445), running services, and the detected Windows Operating System.

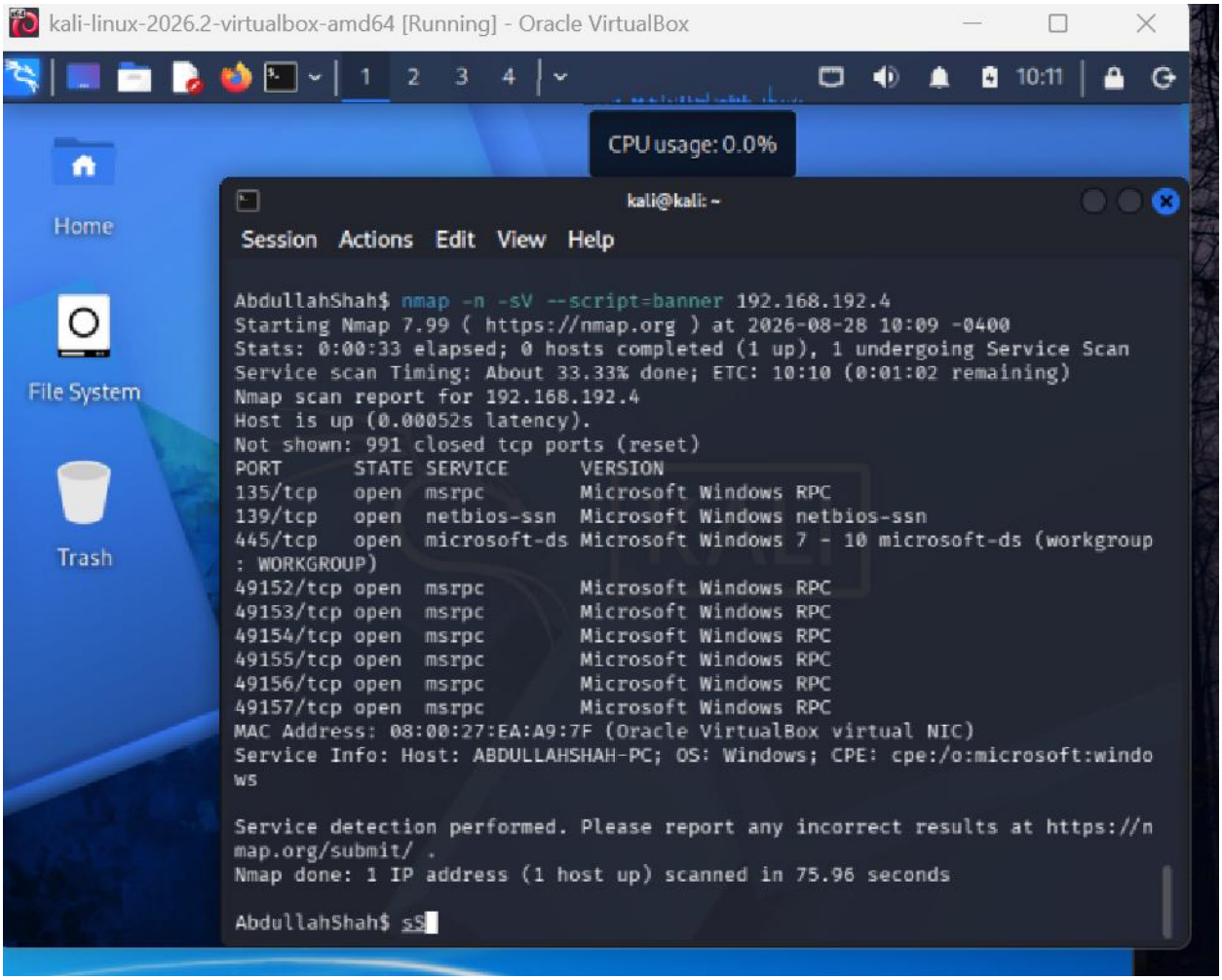




5. Wireshark Traffic Analysis

Wireshark was used to capture packets on the eth0 interface. Display filters were applied to isolate different types of traffic:

- **ARP:** Filtering by arp showed Who-has requests,showing how devices discover MAC addresses on the local network.
- **ICMP:** Filtering by icmp displayed Echo (ping) requests and replies generated during the connectivity testing phase.
- **TCP:** Filtering by tcp showed the transmission control protocol handshakes and data transfers occurring between the machines, especially port 50145 and others.
- **ICMPv6:** Filtering by icmpv6 showed Multicast Listener Reports running in the background.

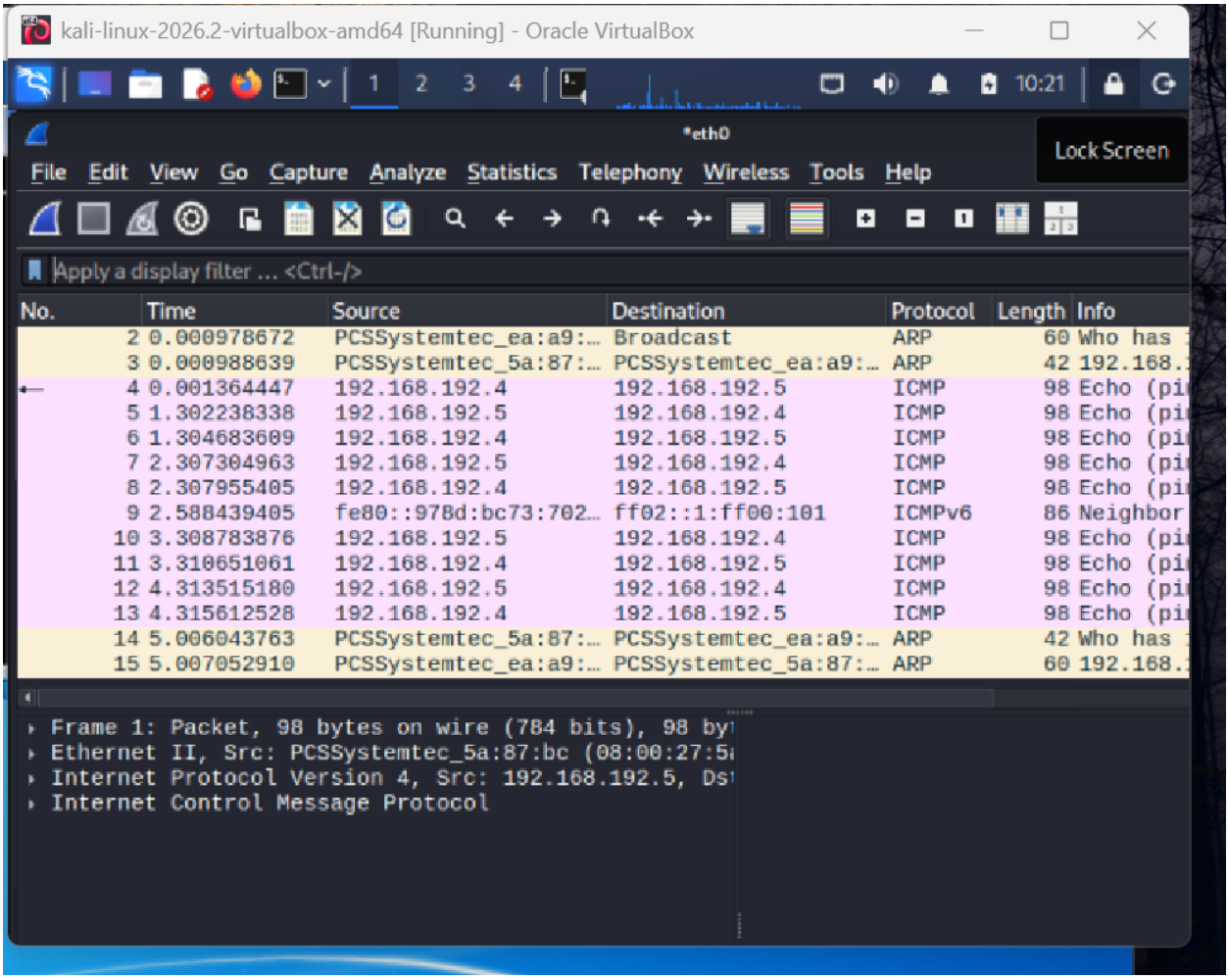


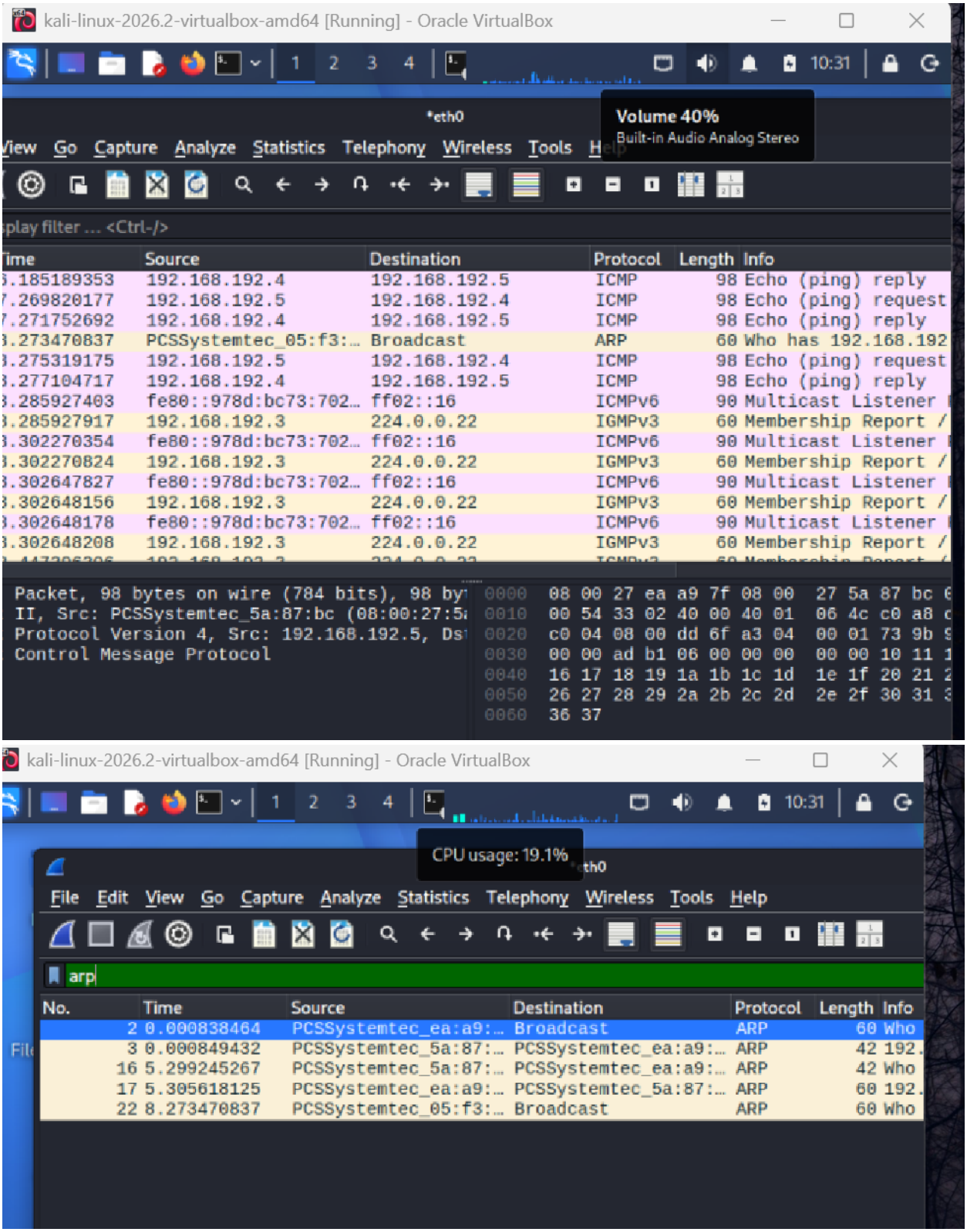

```
AbdullahShah$ nmap -n -sV -p 139,445 192.168.192.4
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-28 10:17 -0400
Nmap scan report for 192.168.192.4
Host is up (0.0014s latency).

PORT    STATE SERVICE      VERSION
139/tcp open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp open  microsoft-ds Microsoft Windows 7 - 10 microsoft-ds (workgroup:
WORKGROUP)
MAC Address: 08:00:27:EA:A9:7F (Oracle VirtualBox virtual NIC)
Service Info: Host: ABDULLAHSHAH-PC; OS: Windows; CPE: cpe:/o:microsoft:windo
ws

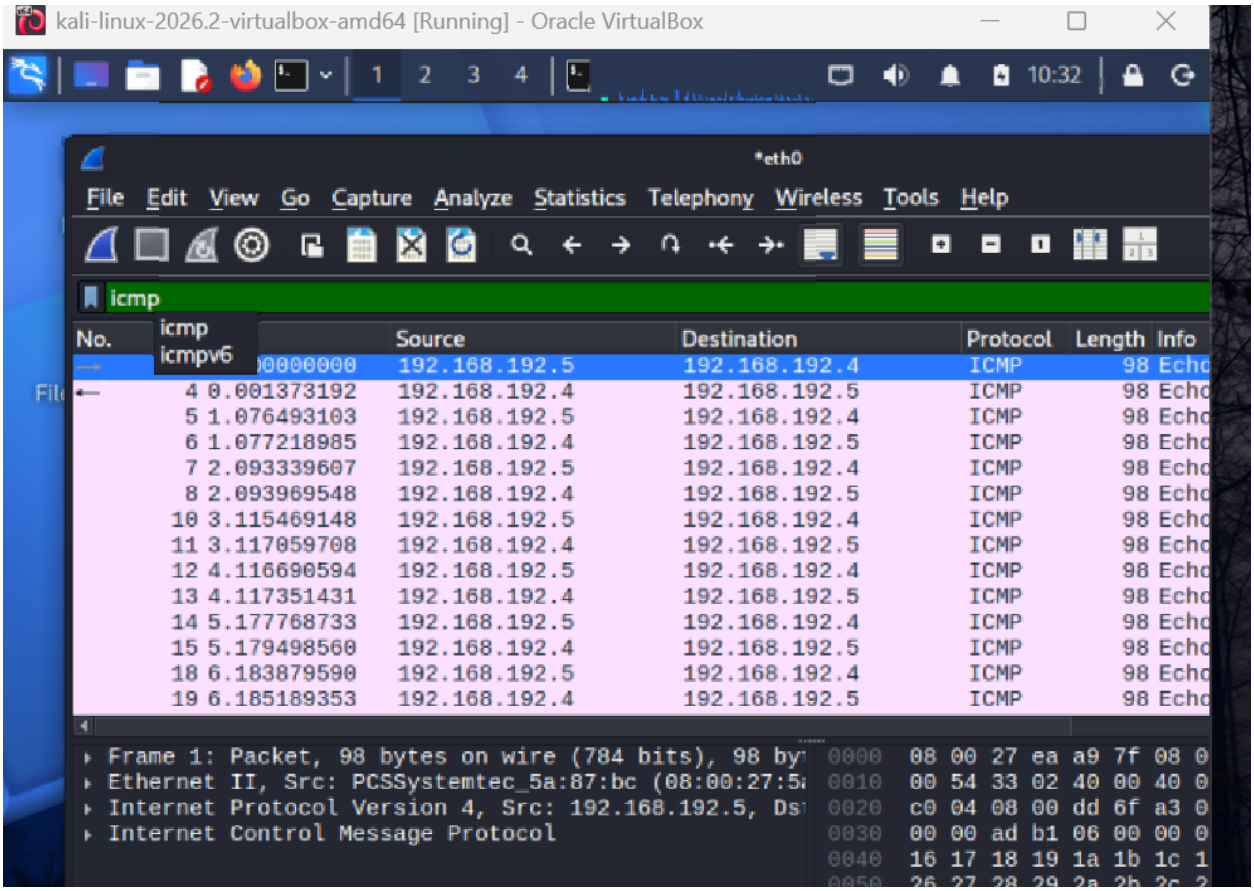
Service detection performed. Please report any incorrect results at https://n
map.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.47 seconds

AbdullahShah$
```

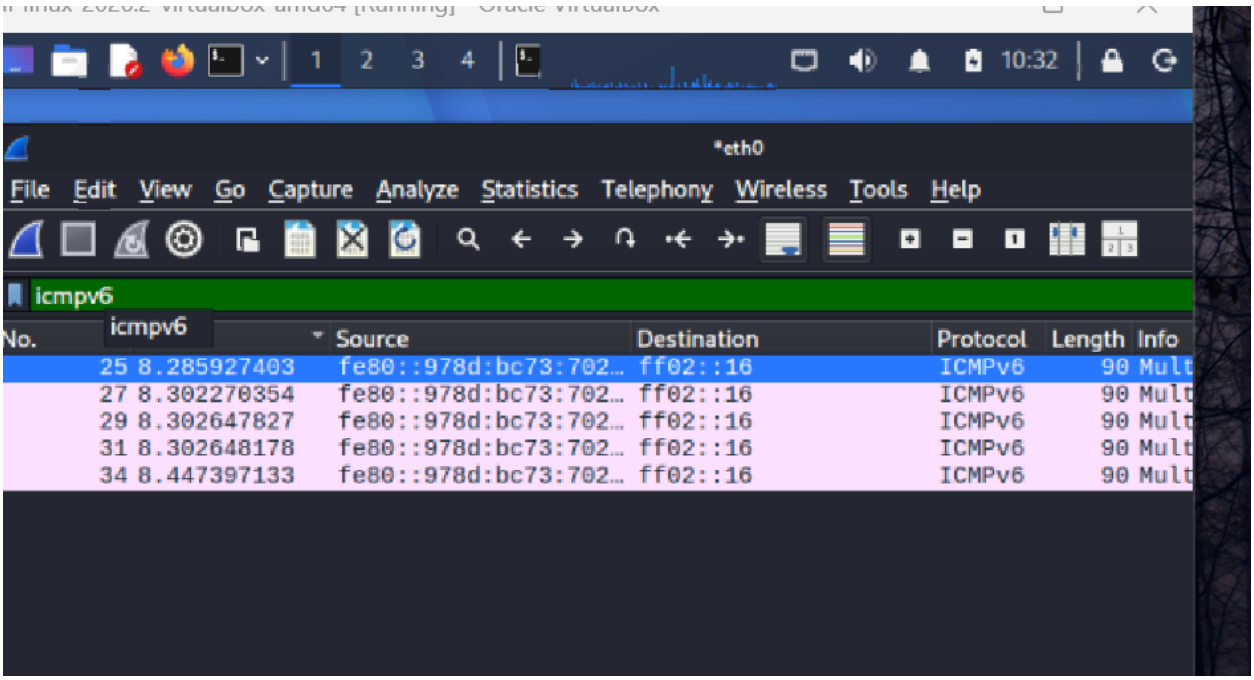

kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
10:21
*eth0
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Lock Screen
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
2 0.000978672 PCSSystemtec_ea:a9:... Broadcast ARP 60 Who has
3 0.000988639 PCSSystemtec_5a:87:... PCSSystemtec_ea:a9:... ARP 42 192.168.
4 0.001364447 192.168.192.4 192.168.192.5 ICMP 98 Echo (pi
5 1.302238338 192.168.192.5 192.168.192.4 ICMP 98 Echo (pi
6 1.304683609 192.168.192.4 192.168.192.5 ICMP 98 Echo (pi
7 2.307304963 192.168.192.5 192.168.192.4 ICMP 98 Echo (pi
8 2.307955405 192.168.192.4 192.168.192.5 ICMP 98 Echo (pi
9 2.588439405 fe80::978d:bc73:702... ff02::1:ff00:101 ICMPv6 86 Neighbor
10 3.308783876 192.168.192.5 192.168.192.4 ICMP 98 Echo (pi
11 3.310651061 192.168.192.4 192.168.192.5 ICMP 98 Echo (pi
12 4.313515180 192.168.192.5 192.168.192.4 ICMP 98 Echo (pi
13 4.315612528 192.168.192.4 192.168.192.5 ICMP 98 Echo (pi
14 5.006043763 PCSSystemtec_5a:87:... PCSSystemtec_ea:a9:... ARP 42 Who has
15 5.007052910 PCSSystemtec_ea:a9:... PCSSystemtec_5a:87:... ARP 60 192.168.
Frame 1: Packet, 98 bytes on wire (784 bits), 98 by
Ethernet II, Src: PCSSystemtec_5a:87:bc (08:00:27:5
Internet Protocol Version 4, Src: 192.168.192.5, Ds
Internet Control Message Protocol

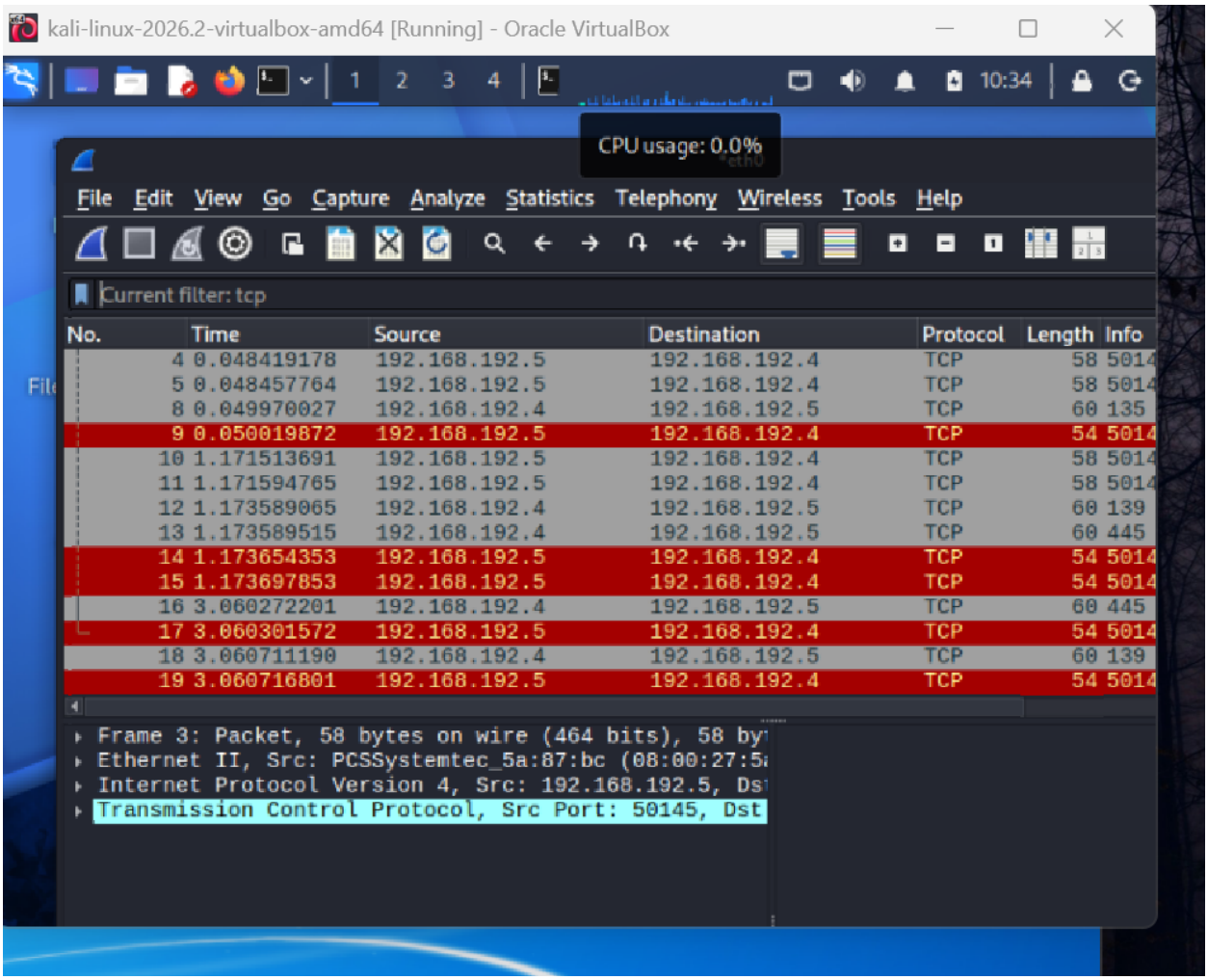


Wireshark capture filtered for arp, showing local network physical address resolution.



Wireshark capture filtered for icmp, displaying ping request and reply packets.





Wireshark capture filtered for tcp, showing active TCP connections and port communications.

6. Vulnerability Identification

Based on the Nmap scans, a critical vulnerability was identified on the target machine.

- **Weakness:** The Windows 7 machine has Port 445 (microsoft-ds / SMB) open and exposed to the network.
- **Risk & Impact:** Port 445 is used for Server Message Block (SMB) file sharing. On an outdated operating system like Windows 7, this specific open port shows suspicion towards the MS17-010 (EternalBlue) exploit. If exploited, it allows an unauthorized attacker to execute arbitrary code remotely and gain full administrative compromise of the target system.
- During scanning, it was also observed that the Windows 7 Firewall was initially blocking the scan, causing all ports to appear as "filtered" with no response. After temporarily disabling the firewall for testing purposes, the same scan revealed the actual open ports and

services. This confirmed that a properly configured firewall can act as an effective first layer of defense against network reconnaissance.

7. Conclusion & Security Improvements The isolated lab setup successfully demonstrated network discovery, traffic analysis, and vulnerability identification without interacting with outside public networks. The presence of an exposed and outdated SMB service (Port 445) shows security risk.

Recommendations:

1. Apply network firewall rules to block inbound traffic on port 445 from untrusted IP addresses.
2. Apply for the latest Microsoft security patches (specifically for MS17-010).
3. Disable SMBv1 on the Windows 7 machine if legacy support is not strictly required.

Q2

First of all I installed dvwa and apache2 and mysql in kali linux using commands sudo apt update

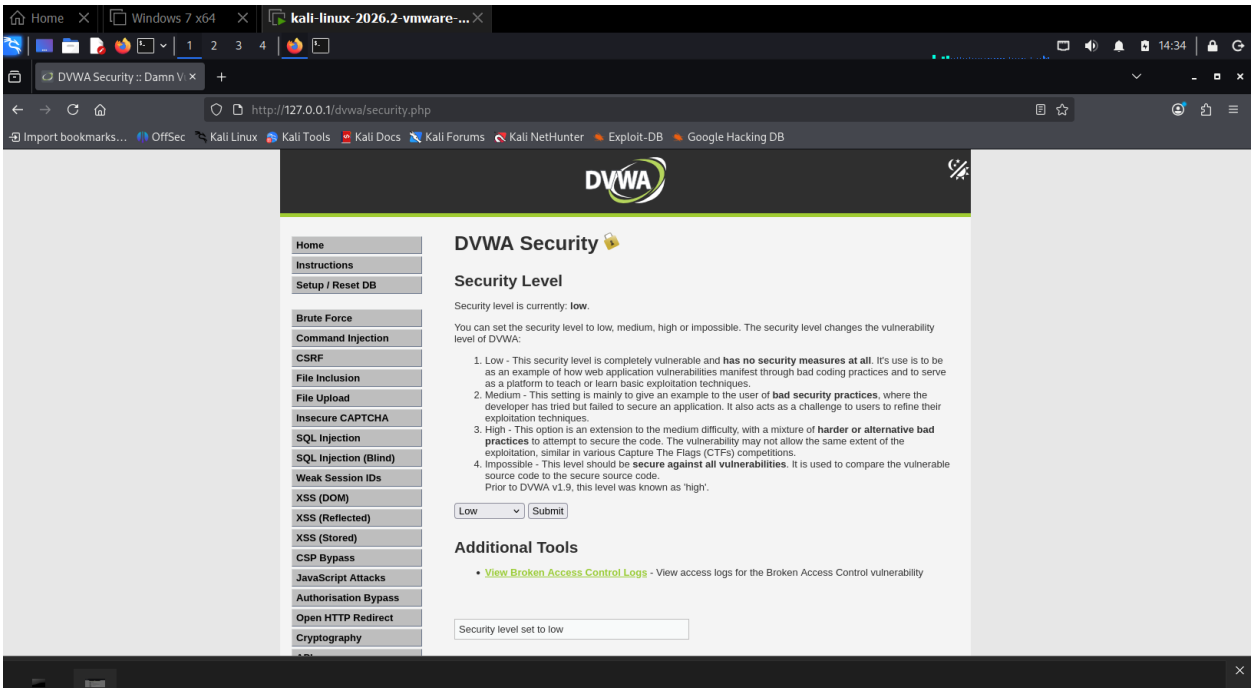
sudo apt install dvwa -y

sudo service apache2 start

sudo service mysql start

DVWA Module	**OWASP Top 10 Category**
SQL Injection	A03:2021 – Injection
Command Injection	A03:2021 – Injection
XSS (Reflected/Stored)	A03:2021 – Injection (XSS)
CSRF	A01:2021 – Broken Access Control
File Upload	A05:2021 – Security Misconfiguration

My DVWA is working and I set Level to low.

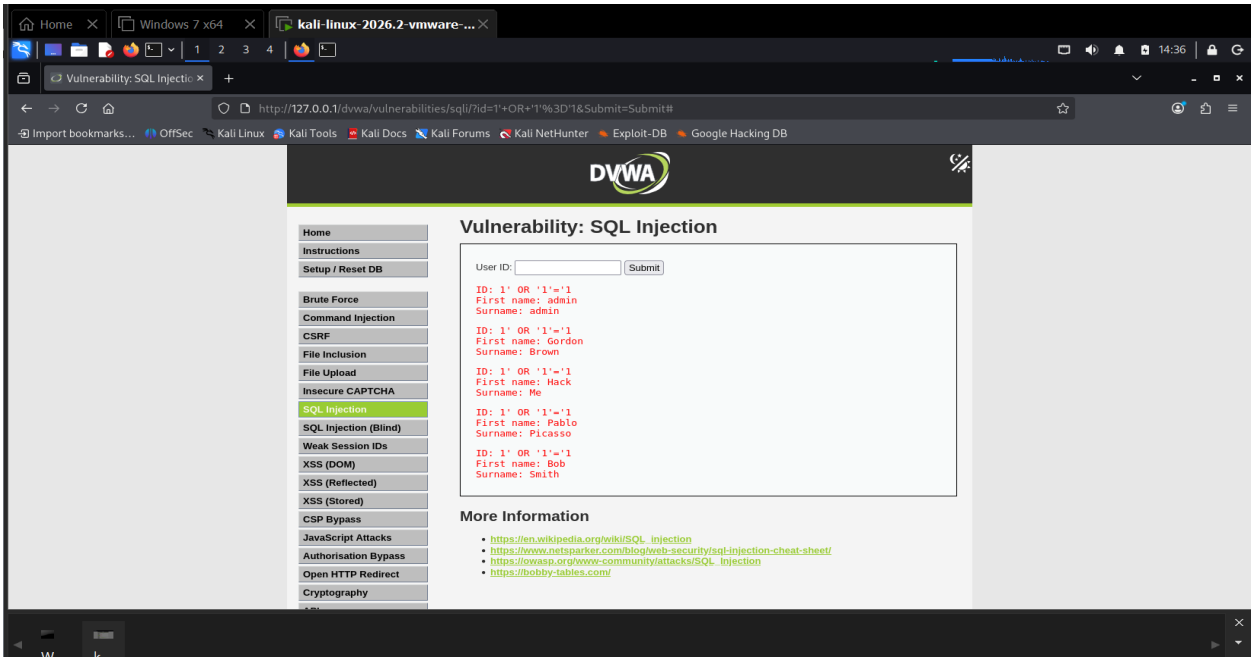


Trying Few SQL Injections.

1' OR '1'='1
%' OR '1'='1

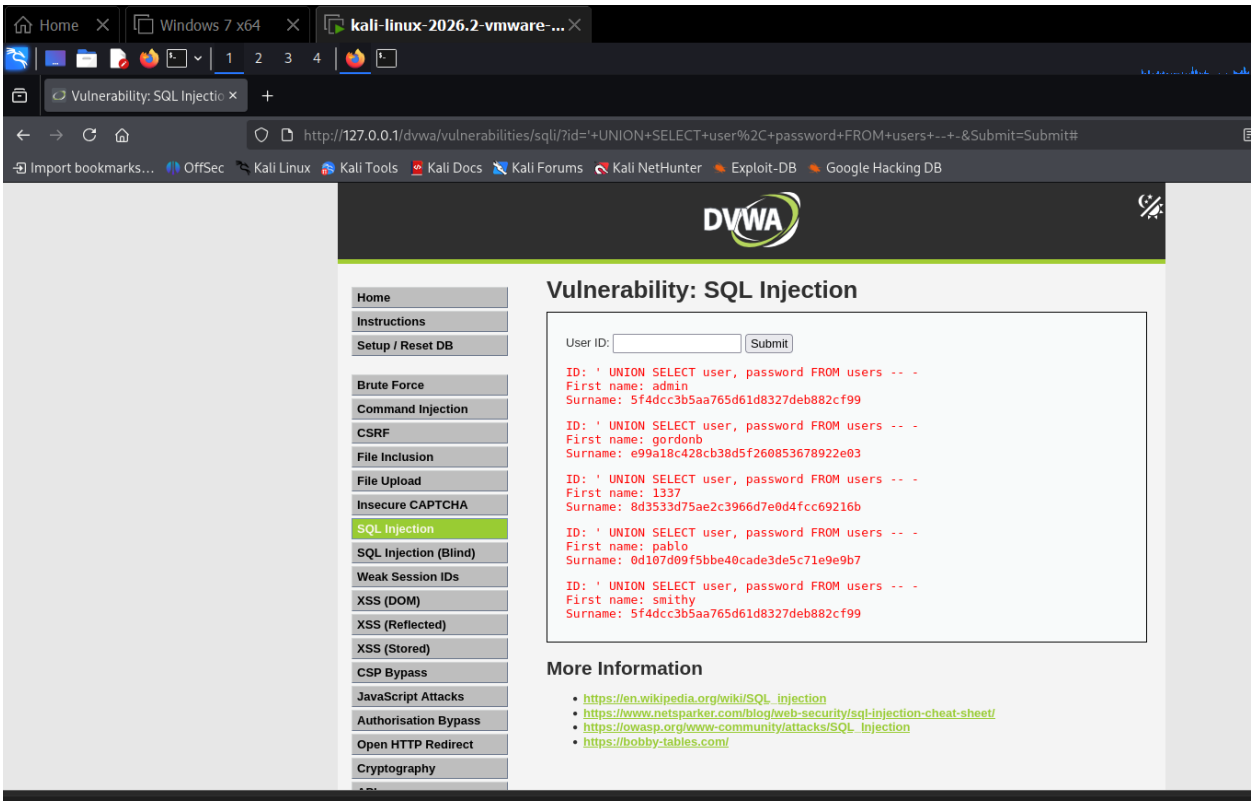
' UNION SELECT user, password FROM users -- -

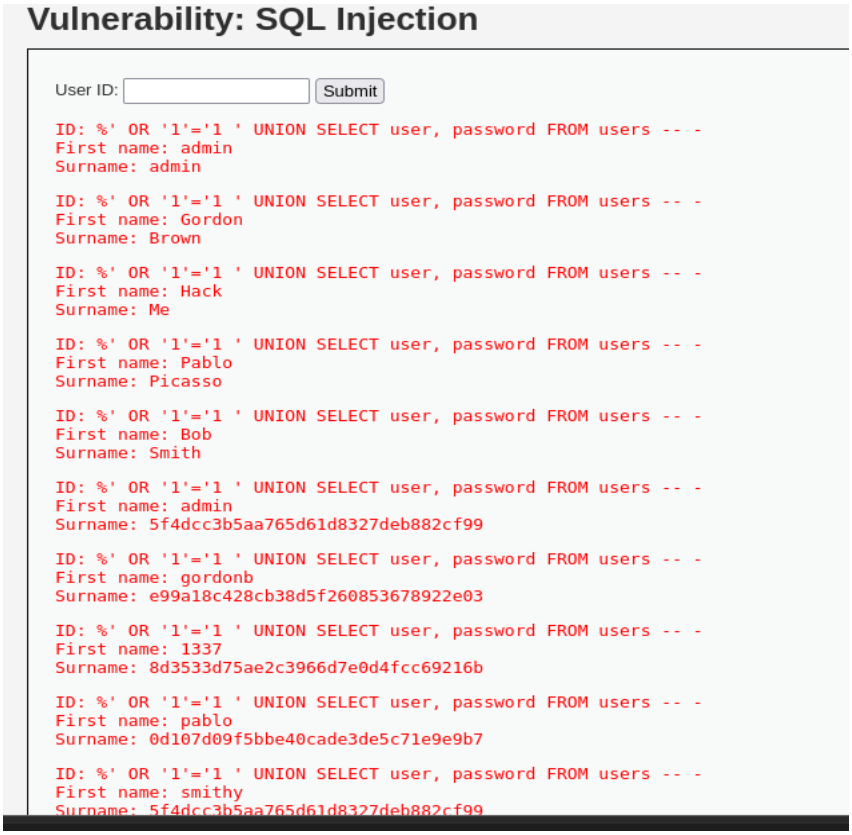
Because user input is concatenated directly into the SQL query without sanitization, an attacker can alter the query's logic to bypass authentication or extract data they shouldn't have access to (e.g., other users' credentials), which could lead to full account compromise or database takeover.



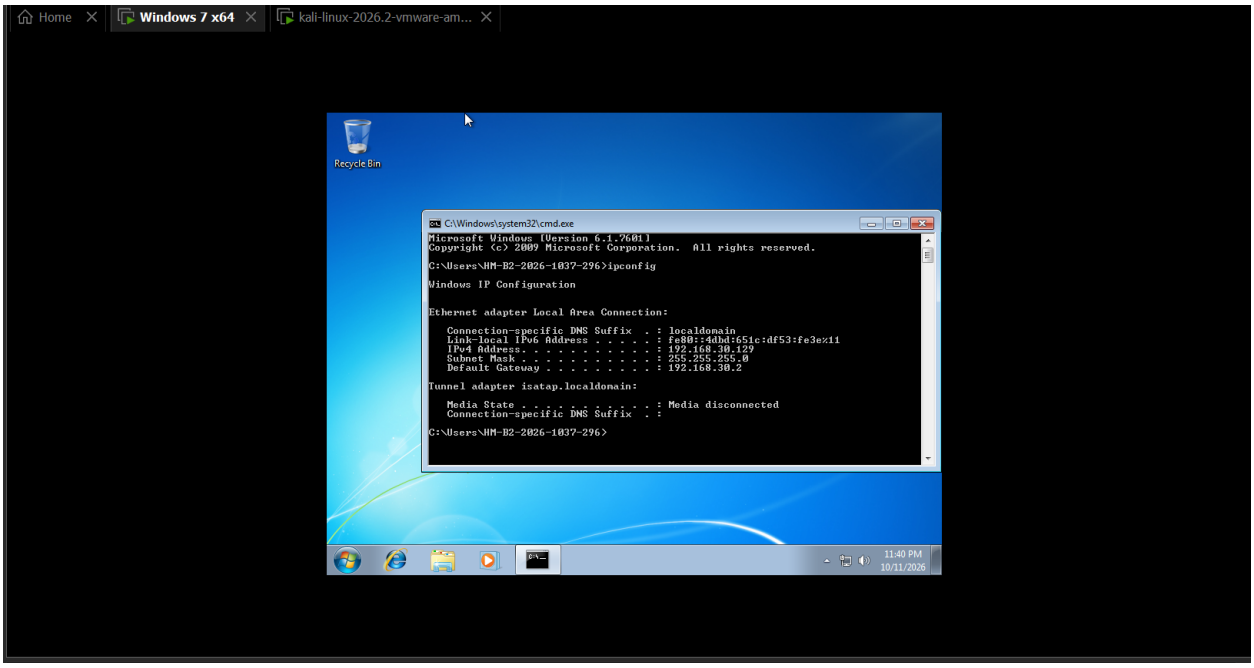


With SQL injection I got the output which I was not supposed to see.

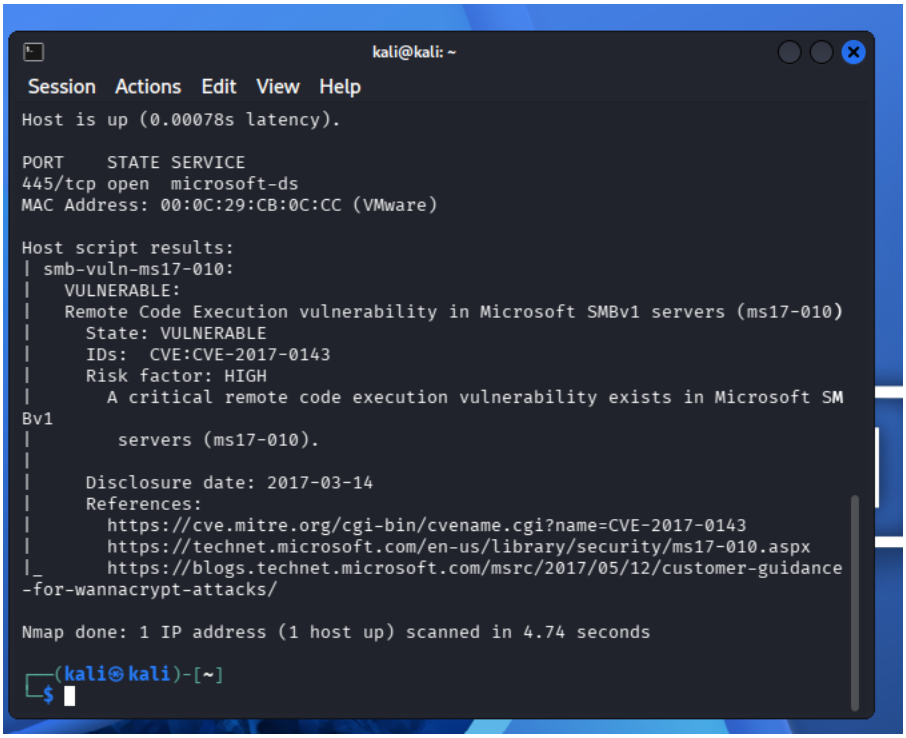


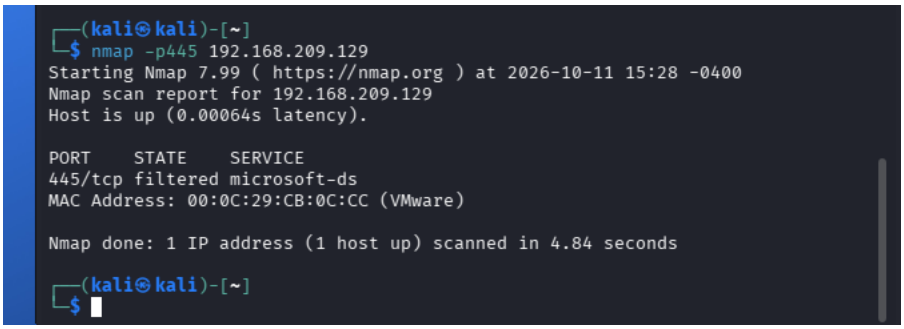


I am checking the IP of my Windows machine.



Port 445 is showed open because I turned OFF firewall of windows.



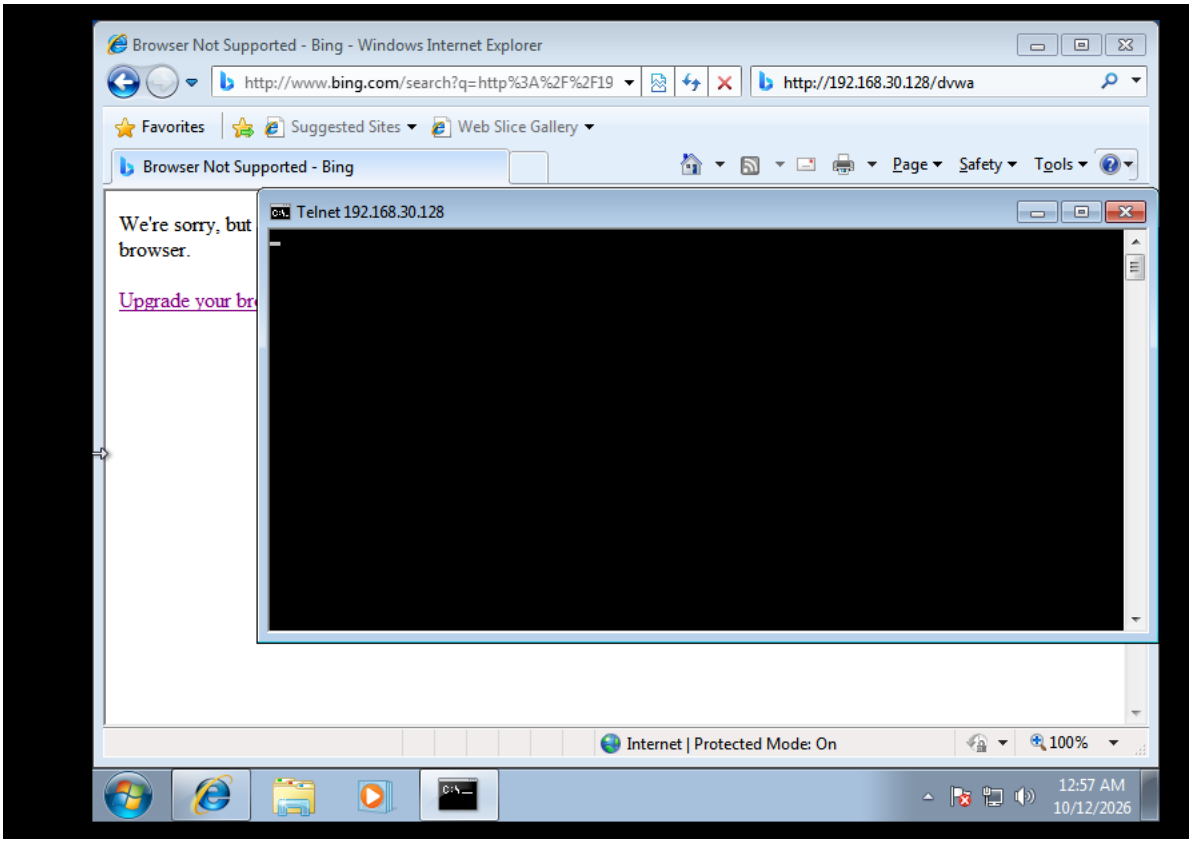


I turned ON firewall of windows.

Port 445/tcp shows that state is Filtered.

Firewall Protection (DVWA Access Control)

To demonstrate a protective firewall measure, a ufw rule was applied on the Kali machine to block the Windows 7 machine from accessing the DVWA web service on port 80.



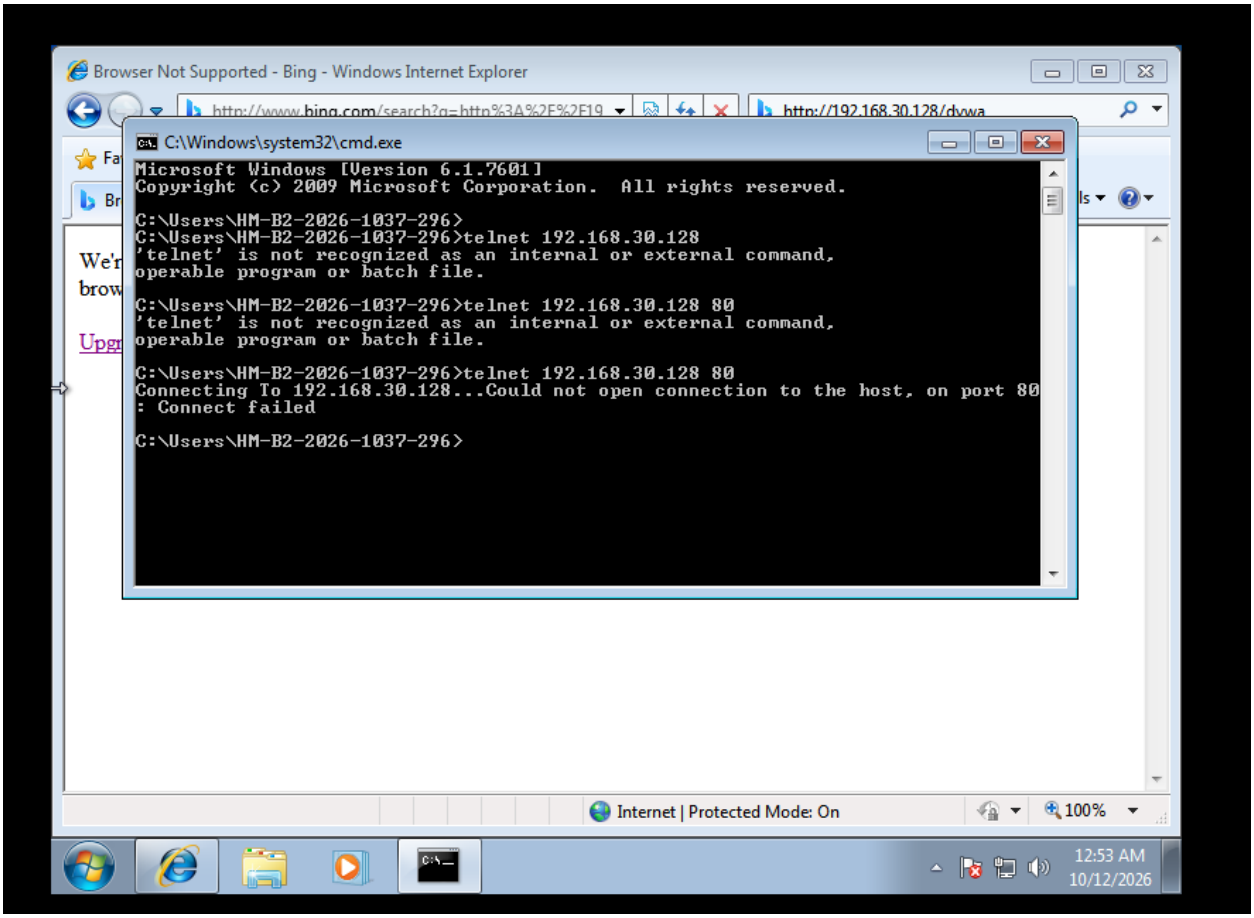
Telnet test from Windows 7 to DVWA (port 80) before applying the firewall rule, showing a successful connection (blank screen indicates open socket).

```
┌──(kali㉿kali)-[~]
└─$ sudo ufw enable
sudo ufw deny from 192.168.30.129 to any port 80
sudo ufw status numbered
Firewall is active and enabled on system startup
Rule added
Status: active

     To                         Action      From
     --                         ------      ----
[ 1] 80                         DENY IN     192.168.30.128
[ 2] 80                         DENY IN     192.168.30.129


┌──(kali㉿kali)-[~]
└─$
```

Firewall rule applied on Kali using ufw, blocking inbound traffic on port 80 from the Windows 7 IP (192.168.30.129).



Telnet test repeated after the firewall rule was applied connection failed, confirming port 80 is now blocked from Windows 7.

Security Verification

Before applying the firewall rule, a telnet connection from Windows 7 to Kali on port 80 succeeded, confirming DVWA was openly accessible. After applying the rule sudo ufw deny from 192.168.30.129 to any port 80, the same telnet attempt failed with “Connect failed,” confirming that the firewall rule successfully restricted access to the web service from that machine.

Digital Forensics Overview

After finishing the penetration test, we need to save all our proof safely. This helps show exactly what happened and supports our work:

- Timestamps: The exact date and time of everything we did to make a clear timeline.
- Screenshots: Pictures of the vulnerabilities, scan results, and firewall changes.
- IP Addresses: The IPs of our attacking machine (Kali) and target machine (Windows 7) to show which computers were involved.
- Nmap Logs: Saved files showing open ports and network weaknesses.
- Wireshark Captures: Records of all the network traffic during our test.
- Web Server Logs: Records showing the web requests made during our attacks.
- Database Records: The sensitive data we were able to see, like passwords or usernames.

Saving all this proves our findings are real and keeps a solid record just in case anyone needs to check our work.

Report Summary

Title: Web Application Penetration Test for DVWA and Windows 7 Lab

Scope: We only tested a private lab with two machines: Kali Linux and Windows 7. We didn't scan or attack anything outside this safe lab.

Evidence: We saved screenshots of our attacks, Nmap scans showing open ports, and tests showing the network before and after the firewall was turned on.

Vulnerabilities Found:

1. SQL Injection in DVWA: The website takes user input without checking it properly, allowing us to mess with the database.
2. Exposed SMB Service: Windows 7 had port 445 open, which has a famous and dangerous vulnerability called MS17-010 (EternalBlue).

Risk Level: Both are high risk. SQL Injection is very common and easy for attackers to use. The Windows 7 flaw is also very dangerous and has been used in huge real-world attacks like WannaCry.

Impact: SQL Injection lets hackers steal passwords or bypass the login screen completely. The Windows flaw lets hackers take full control of the Windows 7 computer remotely.

Recommendations to Fix This:

- Write safer code using parameterized queries so hackers can't inject bad SQL commands.
- Update Windows 7 to patch the flaw or just turn off the old SMBv1 service.
- Use firewalls to only allow trusted computers to access important services.
- Always test the systems again after fixing them to make sure they are safe.

Final Conclusion: We successfully found two major weak spots in our lab: a web flaw (SQL Injection) and a network flaw (SMB on Windows 7). We also showed how a firewall can block access to protect the network. This proves exactly why we need secure coding, regular updates, and strong firewalls to stay safe.